

Criminal Eye

Professional Development Roadmap

AI-Powered Forensic Sketch Generation & Criminal Face Recognition System

Purpose

This roadmap provides a structured, industry-style implementation plan for building the Criminal Eye platform from initial project setup to deployment. Each phase produces a working milestone before introducing additional complexity.

Phase 1 – Project Foundation

- Create a Next.js 16 project with TypeScript, App Router, Tailwind CSS v4, ESLint, and Turbopack.
- Remove boilerplate files and establish a clean architecture.
- Create folders: app, components, features, hooks, lib, actions, services, schemas, types, utils, config, constants, styles.
- Install core libraries: shadcn/ui, Lucide React, Motion, React Hook Form, Zod, Sonner, next-themes, clsx, class-variance-authority, tailwind-merge, date-fns.

Phase 2 – Database Layer

- Provision a PostgreSQL database using Neon or Supabase.
- Configure Prisma ORM and migrations.
- Design entities: User, Case, Witness, Sketch, Criminal, Recognition Result, Report, Audit Log.

Phase 3 – Authentication & Security

- Integrate Better Auth.
- Implement login, registration, sessions, protected routes, and role-based access (Admin, Investigator, Officer).

Phase 4 – UI System

- Build the design system (buttons, forms, dialogs, tables, cards).
- Create responsive dashboard layout with sidebar and navigation.

Phase 5 – Business Modules

- Develop Case Management CRUD.
- Develop Criminal Database CRUD.
- Integrate secure image storage with AWS S3 or Supabase Storage.

Phase 6 – AI Backend

- Create a separate FastAPI service.
- Expose REST APIs for witness processing, sketch generation, and face recognition.

Phase 7 – AI Features

- Convert witness descriptions into structured facial attributes using an LLM.
- Generate sketches using a diffusion model.
- Generate facial embeddings using InsightFace.
- Perform vector similarity search with pgvector.
- Return ranked suspect matches with confidence scores.

Phase 8 – Reporting & Deployment

- Generate downloadable PDF investigation reports.
- Test authentication, APIs, AI pipeline, and database.
- Deploy frontend on Vercel, AI service on Railway/Render, PostgreSQL on Neon, storage on AWS S3.

Recommended Milestones

Milestone 1: Foundation (Next.js, UI, Database, Authentication)

Milestone 2: Business Modules (Cases, Criminal Database, Storage)

Milestone 3: AI Integration (LLM, Sketch Generation, Recognition)

Milestone 4: Production (Reports, Testing, Deployment, Documentation)

Expected Outcome

At the end of this roadmap, the project will be a production-style full-stack AI web application with a modular architecture, secure authentication, scalable database, dedicated AI service, automated sketch generation, intelligent suspect matching, and deployment-ready infrastructure.